Lecture 7

Smooth manifolds

Objectives. *A smooth manifold has local coordinates whose changes are smooth. This allows derivatives to be compared between coordinate systems. We construct coordinates on spheres and projective spaces, then prove that regular constraints define manifolds using the implicit function theorem. The space of pure qubit states gives a concrete example: a sphere described by two complex coordinate charts.*

By the end of this lecture you will be able to:

- Check chart transitions and construct a smooth atlas.
- Test smooth maps in coordinates and apply the chain rule.
- Prove that a nonempty regular level set is an embedded submanifold.
- Relate the two projective qubit coordinates by $w = 1/z$.

Unit 7A Charts and the complex projective line

7.1 Charts and smooth structures

Definition 7.1 (Topological manifold). A *topological n-manifold* is a Hausdorff, second-countable space M locally homeomorphic to open subsets of $\mathbb{R}^n$. Here second-countable means that its topology has a countable basis.

We consider manifolds without boundary.

Definition 7.2 (Chart). A *chart* is a pair (U, φ), with $U \subseteq M$ open and $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ a homeomorphism onto an open subset. The component functions of the chart are its *coordinates*, written $x^1, \dots, x^n$. The superscripts are coordinate labels, not powers.

By Lecture 1, equivalent norms give the same open sets. For $n \geq 1$, the Euclidean and maximum norms on $\mathbb{R}^n$ are equivalent. The sum of the n squared coordinates is at least the largest one and at most n times it. Taking square roots gives

$$\|v\|_\infty \leq \|v\|_2 \leq \sqrt{n}\,\|v\|_\infty \qquad (v \in \mathbb{R}^n).$$

Thus the meaning of an open chart image does not depend on which of these norms is used.

On the right semicircle, the angle θ and the vertical coordinate y describe the same point by

$$(\cos\theta, \sin\theta), \qquad -\pi/2 < \theta < \pi/2, \qquad y = \sin\theta.$$

The transition has the smooth inverse $\theta = \arcsin y$ on $(-1, 1)$. For the scalar function $f(\cos\theta, \sin\theta) = \theta$, the angle expression has derivative 1. Its expression in the vertical

coordinate is $\arcsin y$, whose derivative is $1/\sqrt{1-y^2}$. The chain rule relates these derivatives because $\mathrm{d}y/\mathrm{d}\theta = \cos\theta$. The marked point in the drawing has angle $\pi/6$ and vertical coordinate $1/2$.

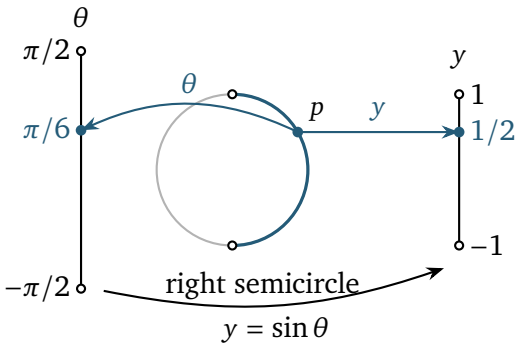


Figure 7.1: The same point has coordinates $\pi/6$ and $1/2$. The transition $y = \sin\theta$ compares coordinates on open intervals.

For a function $f\colon M \to \mathbb{R}$, its expression in a chart is $f \circ \varphi^{-1}$. On a second chart the expression is

$$f \circ \psi^{-1} = (f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1}).$$

Smooth transitions make smoothness of the function independent of its coordinate description, by the Euclidean chain rule.

Definition 7.3 (Compatible charts). Two charts are *smoothly compatible* when both coordinate changes on their overlap are smooth. For charts (U, φ) and (V, ψ), one such change is

$$\psi \circ \varphi^{-1}\colon \varphi(U \cap V) \longrightarrow \psi(U \cap V).$$

Smooth means C^∞: all orders of partial derivatives exist and are continuous. An empty overlap imposes no condition.

The charts describe the same points of $U \cap V$. The transition compares their descriptions between open subsets of $\mathbb{R}^n$. The drawing uses $M = \mathbb{R}^2$, the rectangles $U = (-2, 1) \times (-1, 1)$ and $V = (-1, 2) \times (-1, 1)$, and charts $\varphi(x, y) = (x, y)$ and $\psi(x, y) = (x + 2, y)$.

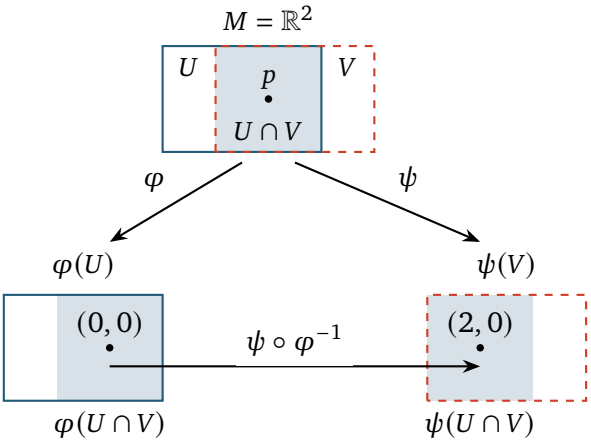


Figure 7.2: Only the shaded overlap images are the source and target of $\psi \circ \varphi^{-1}$. The marked point $p = (0, 0)$ has coordinates $(0, 0)$ and $(2, 0)$.

Definition 7.4 (Smooth atlas)**.** A *smooth atlas* is a collection of pairwise compatible charts covering M.

Example 7.5 (An angle atlas on the circle)**.** Take $U = S^1 \setminus \{(1,0)\}$ with angle $\theta \in (0, 2\pi)$. Take $V = S^1 \setminus \{(-1,0)\}$ with angle $\theta' \in (-\pi, \pi)$. The overlap has an upper arc and a lower arc. On these arcs the transition is respectively

$$\theta' = \theta \quad (0 < \theta < \pi), \qquad \theta' = \theta - 2\pi \quad (\pi < \theta < 2\pi).$$

Both transitions and their inverses are smooth, so these charts form an atlas. A single chart cannot cover the circle. Its image would be a nonempty compact open subset of $\mathbb{R}$. Compactness makes that image closed and bounded. Connectedness of $\mathbb{R}$ would then make the image all of $\mathbb{R}$, contradicting boundedness.

Definition 7.6 (Smooth structure)**.** A *smooth structure* is a maximal smooth atlas, containing every chart compatible with it.

Definition 7.7 (Smooth manifold)**.** A *smooth manifold* is a topological manifold with a smooth structure.

A covering atlas specifies which coordinate descriptions are compatible. Its maximal extension includes every chart compatible with that choice.

Any smooth atlas determines a unique smooth structure. To prove this, adjoin every chart compatible with the given atlas. For two added charts ψ and θ and a point in their overlap, choose an original chart φ containing that point. Locally the transition is

$$\theta \circ \psi^{-1} = (\theta \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1}).$$

Both factors are smooth. Reversing the roles gives a smooth inverse. Thus the enlarged collection is an atlas. Any chart compatible with the enlarged collection was compatible with the original one and has already been added. This proves maximality. Every chart of any maximal atlas containing the original atlas belongs to this enlarged collection. That maximal atlas must therefore equal the enlarged collection, proving uniqueness.

The smooth structure is extra data. On the topological line, the global charts $x(t) = t$ and $y(t) = t^3$ individually define smooth structures. Their transition $y \circ x^{-1}(s) = s^3$ is smooth, but its inverse is not differentiable at zero. Along $s = 1/k^3$, its difference quotient is $(1/k)/(1/k^3) = k^2$, which is unbounded as $k \to \infty$. Hence the two charts do not belong to one smooth atlas. The function $g(t) = t$ has expression $g \circ x^{-1}(s) = s$ in the first structure and $g \circ y^{-1}(s) = \sqrt[3]{s}$ in the second. Thus the two structures give different sets of smooth real-valued functions. We compare these structures by a smooth map in §7.3.

Local Euclidean coordinates alone do not imply Hausdorffness. Identify two copies of $\mathbb{R}$ at every nonzero coordinate, leaving two origins. Each origin has an interval chart, but any two neighbourhoods of the origins intersect. The sequence with coordinates $1/k$, for positive integers k, converges to both origins. Hausdorffness excludes this failure of uniqueness of limits. The drawing separates the two copies vertically only to distinguish their origins.

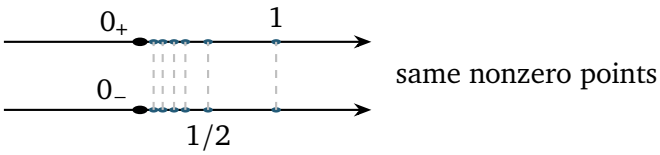


Figure 7.3: Nonzero points in the two rows are identified. Both origins are limits of the same positive sequence, so the quotient is not Hausdorff.

An uncountable disjoint union of lines is locally Euclidean and Hausdorff. It is not second-countable: each component would need a different nonempty basis element contained in it. Second countability supplies the countable chart constructions used for partitions of unity and integration in Lecture 12.

A smooth invertible transition also fixes the dimension. By the chain rule, its derivative has a two-sided inverse, so its source and target have the same dimension.

7.2 Spheres, projective spaces and products

Write a point of the unit sphere $S^n \subset \mathbb{R}^{n+1}$ as (y, t), where $y \in \mathbb{R}^n$ and $t \in \mathbb{R}$. Let $N = (0, 1)$ and $S = (0, -1)$ be its north and south poles. Stereographic projection gives charts

$$\varphi_N(y,t) = \frac{y}{1-t}, \qquad \varphi_S(y,t) = \frac{y}{1+t}$$

on $S^n \setminus \{N\}$ and $S^n \setminus \{S\}$ respectively. Both have image $\mathbb{R}^n$. The projection is shown in a two-dimensional slice. The transition $u \mapsto u/|u|^2$ is inversion in the unit sphere: the two coordinates shown have lengths 2 and $1/2$.

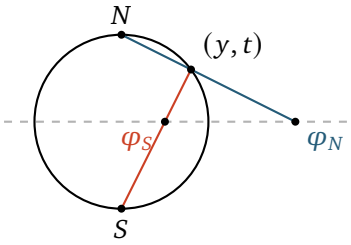


Figure 7.4: The ray from the omitted pole meets the equatorial coordinate plane. The two projections describe the same sphere away from their respective poles.

A line through zero in $\mathbb{R}^3$ with $X^0 \neq 0$ meets the plane $X^0 = 1$ at exactly one point. Its representative there is $(1, u^1, u^2)$, where $(u^1, u^2) = (X^1/X^0, X^2/X^0)$. The drawing shows these ratios as affine coordinates of the intersection.

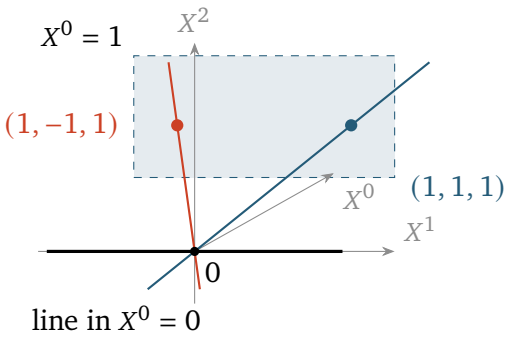


Figure 7.5: Two lines meet the chart plane $X^0 = 1$. The line in $X^0 = 0$ misses that plane and is outside this chart.

On the overlap with $X^1 \neq 0$, we divide the representative by u^1:

$$(1, u^1, u^2) \sim (1/u^1, 1, u^2/u^1), \qquad \varphi_1 \circ \varphi_0^{-1}(u^1, u^2) = (1/u^1, u^2/u^1), \quad u^1 \neq 0.$$

This transition and its inverse are smooth rational functions. For $\mathbb{R}\mathrm{P}^1$ the transition is $u \mapsto 1/u$. This is also the stereographic transition on S^1.

Proposition 7.8 (Standard atlases). *For $n \geq 1$, the sphere S^n and real projective space $\mathbb{R}\mathrm{P}^n$ are smooth n-manifolds. Products of smooth manifolds have smooth product atlases. In particular, $T^2 = S^1 \times S^1$ is a smooth surface.*

Proof. First, the inverse north chart is

$$\varphi_N^{-1}(u) = \left(\frac{2u}{1 + |u|^2}, \frac{|u|^2 - 1}{1 + |u|^2} \right).$$

Its squared norm is one. Substitution in $y/(1 - t)$ gives u. Conversely, for $(y, t) \neq N$, the relation $|y|^2 = 1 - t^2$ gives $|y/(1 - t)|^2 = (1 + t)/(1 - t)$. Substituting this into the displayed formula returns (y, t). Thus it is the continuous inverse. The inverse south chart changes the sign of the last component. On the overlap, substitution gives $\varphi_S \circ \varphi_N^{-1}(u) = u/|u|^2$ for $u \neq 0$. This map is smooth and is its own inverse. The sphere is Hausdorff and second-countable as a subspace of Euclidean space. These charts therefore define its smooth structure.

Next, use Lecture 2's quotient $q \colon S^n \to \mathbb{R}\mathrm{P}^n$ by antipodes. The map q is open: the inverse image of the image of an open set W is $W \cup (-W)$. Distinct two-point orbits have disjoint small antipodally invariant neighbourhoods, proving Hausdorffness of the quotient. Images of a countable sphere basis give a countable quotient basis.

Write a class of unit vectors as $[X^0 : \cdots : X^n]$. A nonzero vector represents the class of its normalisation. Multiplying it by a nonzero scalar leaves the same antipodal class, so the notation also describes its line through zero. The open patch U_i consists of classes with $X^i \neq 0$. Its coordinates and inverse are

$$\varphi_i([X]) = (X^0/X^i, \ldots, \widehat{X^i/X^i}, \ldots, X^n/X^i), \qquad \varphi_i^{-1}(u) = [u^0 : \cdots : 1 : \cdots : u^n],$$

with the 1 in position i. The remaining coordinate labels omit i. The hat in this display means that the ith entry is omitted. The ratios are invariant under antipodes and continuous upstairs. Since q is open, its restriction over U_i is a quotient map, so the ratios are continuous on U_i. For the inverse, normalise the displayed representative vector and apply q. These continuous maps are inverse: normalising does not change ratios, and the ratios recover

the original vector up to a nonzero scalar. On $U_i \cap U_j$, divide every component of the representative with ith component 1 by its nonzero jth component, then omit component j. All new coordinates are smooth rational functions with nonzero denominator. The same description with i and j exchanged gives the inverse transition.

The $n+1$ patches cover, completing the projective-space assertion.

Finally, take products of charts. Their coordinate changes are products of smooth maps, so they are smooth. Finite products preserve Hausdorffness and second countability: products of the respective basis elements give a countable basis. This proves the product assertion and the torus case. ∎

7.3 Smooth maps and the chain rule

Let M and N be smooth manifolds of dimensions n and m.

Definition 7.9 (Smooth map)**.** A continuous map $f\colon M \to N$ is *smooth* if, near every $p \in M$, its coordinate expression is smooth. For charts φ near p and ψ near $f(p)$, this expression is

$$\widehat{f} = \psi \circ f \circ \varphi^{-1}\colon \varphi(U \cap f^{-1}(V)) \longrightarrow \psi(V).$$

The domain is open because f is continuous.

Definition 7.10 (Diffeomorphism)**.** A *diffeomorphism* is a bijective smooth map with a smooth inverse.

We use the following index convention for coordinate calculus. A repeated upper and lower index is summed from 1 to n, unless another range is stated. For a linear functional on $\mathbb{R}^n$ with coefficient row a_i and a vector in $\mathbb{R}^n$ with component column X^i, write $a_i X^i = \sum_{i=1}^{n} a_i X^i$. The index position distinguishes transformation rules. It does not mean that a metric has been used. For example, a linear coordinate change and the corresponding rules are

$$J = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad x' = Jx, \qquad X' = JX, \qquad a' = aJ^{-1}, \qquad a'X' = aX.$$

The last equality follows by cancelling $J^{-1}J$. Lower coordinate labels x_i in some prerequisite notes become x^i here. This relabelling does not raise the index of a linear functional.

Proposition 7.11 (Independence of coordinates)**.** *Smoothness is independent of the chosen charts. Compositions of smooth maps are smooth. Their coordinate Jacobians obey the chain rule.*

Proof. With different charts $\widetilde{\varphi}$ and $\widetilde{\psi}$, the new expression, on its common local domain, is

$$\widetilde{\psi} \circ f \circ \widetilde{\varphi}^{-1} = (\widetilde{\psi} \circ \psi^{-1}) \circ \widehat{f} \circ (\varphi \circ \widetilde{\varphi}^{-1}).$$

The outer maps are smooth transitions. This proves independence at each point. For a further smooth map $g\colon N \to P$, use the same middle chart on N. The coordinate expression of $g \circ f$ is $\widehat{g} \circ \widehat{f}$, and the Euclidean chain rule gives

$$D(\widehat{g} \circ \widehat{f})_x = D\widehat{g}_{\widehat{f}(x)}\, D\widehat{f}_x.$$

Both smoothness of the composition and the Jacobian formula follow. ∎

The hat in $\widehat{f}$ denotes a coordinate expression of a map. Here $D\widehat{f}_x$ denotes the $m \times n$ matrix of first partial derivatives. In component notation, the same formula is

$$\frac{\partial (g \circ f)^a}{\partial x^i} = \frac{\partial g^a}{\partial y^b} \frac{\partial f^b}{\partial x^i},$$

with the repeated index b summed. These are coordinate expressions; Lecture 8 defines the differential without choosing coordinates.

Example 7.12 (Testing a height function)**.** Consider $h\colon S^n \to \mathbb{R}$ given by $h(y,t) = t$. Its expressions in the stereographic charts are

$$h \circ \varphi_N^{-1}(u) = \frac{|u|^2 - 1}{1 + |u|^2}, \qquad h \circ \varphi_S^{-1}(u) = \frac{1 - |u|^2}{1 + |u|^2}.$$

Both are smooth on $\mathbb{R}^n$, since their denominators are positive. The charts cover the sphere, so h is smooth everywhere.

For the two smooth structures on the line above, the map $f(t) = \sqrt[3]{t}$ from the x-structure to the y-structure has coordinate expression $y \circ f \circ x^{-1}(s) = s$. Its inverse has the identity coordinate expression too. Thus it is a diffeomorphism, although the identity map between those structures is not a diffeomorphism.

7.4 The qubit's projective state space

Physical input. A two-level system has orthonormal reference states $|0\rangle$ and $|1\rangle$. A normalised column $\psi = (a, b)^T$ represents $a\,|0\rangle + b\,|1\rangle$, with $|a|^2 + |b|^2 = 1$. Measurement in this basis has probabilities $|a|^2$ and $|b|^2$. For a Hermitian matrix A, the expectation in any nonzero column ψ is

$$\langle A \rangle_\psi = \frac{\psi^\dagger A \psi}{\psi^\dagger \psi}.$$

Pure states are identified under a common phase. The Pauli matrices in the reference basis are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For a spin-1/2 system the physical spin components are $S_a = (\hbar/2)\sigma_a$, for $a \in \{x, y, z\}$.

A common nonzero scalar cancels between the numerator and denominator of every expectation. Normalisation removes its magnitude, leaving the common phase. Relative phase can affect expectations in other bases.

Definition 7.13 (Complex projective line)**.** The *complex projective line* $\mathbb{C}\mathrm{P}^1$ consists of nonzero columns $(a, b) \in \mathbb{C}^2$ modulo multiplication by a common nonzero complex scalar. Its points are written $[a : b]$. It carries the quotient topology from $\mathbb{C}^2 \setminus \{0\}$.

This is the construction of $\mathbb{R}\mathrm{P}^1$ with $\mathbb{C}$ in place of $\mathbb{R}$. The physical pure-state space is this quotient. The patches with $a \neq 0$ and $b \neq 0$ have coordinates

$$z = b/a, \qquad w = a/b.$$

Their inverses are $z \mapsto [1 : z]$ and $w \mapsto [w : 1]$. For an open set W of nonzero columns, the set of all columns equivalent to some point of W is $\bigcup_{\lambda \in \mathbb{C} \setminus \{0\}} \lambda W$. Each scalar multiplication

is a homeomorphism, so this union is open. The quotient projection is therefore open, as for real projective space. Thus the patches are open and the ratios define continuous charts. On the overlap $z \neq 0$, the transition is $w = 1/z$. For $z = u + iv$, this is the real smooth map

$$(u, v) \longmapsto \left(\frac{u}{u^2 + v^2}, \frac{-v}{u^2 + v^2} \right).$$

It is its own inverse. The missing point of either chart is the origin of the other chart. Lecture 20 differentiates this transition to describe vectors attached to the sphere.

For $\psi = (a, b)^T \neq 0$, the Pauli expectation numerators are

$$\psi^\dagger \sigma_x \psi = 2\,\mathrm{Re}(\bar{a}b), \qquad \psi^\dagger \sigma_y \psi = 2\,\mathrm{Im}(\bar{a}b), \qquad \psi^\dagger \sigma_z \psi = |a|^2 - |b|^2.$$

Definition 7.14 (Bloch vector)**.** The *Bloch vector* of a pure qubit state is its vector of Pauli expectations:

$$B \colon \mathbb{C}\mathrm{P}^1 \longrightarrow \mathbb{R}^3, \qquad B([a : b]) = \frac{1}{|a|^2 + |b|^2} \left(2\,\mathrm{Re}(\bar{a}b),\ 2\,\mathrm{Im}(\bar{a}b),\ |a|^2 - |b|^2\right).$$

For a spin-1/2 system this vector is $2\langle \mathbf{S} \rangle / \hbar$. For a general two-level system its components are Pauli expectations, without an identification with spatial directions. The north pole has $z = 0$, and the south pole has $w = 0$. Both coordinates are available away from the poles, with $w = 1/z$. The drawing places the two basis states and one further ray on the unit sphere.

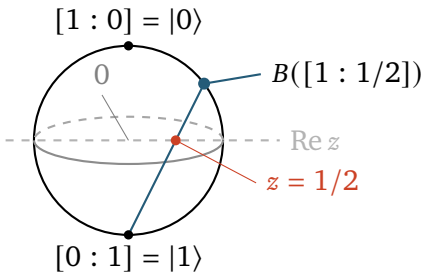


Figure 7.6: The north state $[1 : 0]$ is $|0\rangle$, and the south state $[0 : 1]$ is $|1\rangle$. The ray from the south pole through $B([1 : 1/2]) = (4/5, 0, 3/5)$ meets the coordinate plane at $z = 1/2$.

Example 7.15 (Projective states and the Bloch sphere)**.** We show that the Bloch map takes values in S^2 and is a diffeomorphism. Multiplying a and b by a common nonzero scalar multiplies numerator and denominator by the same positive number, so the map is well-defined. The formula on nonzero columns is continuous with nonvanishing denominator. Lecture 2's quotient criterion therefore makes B continuous. Its squared norm is one because $4|a|^2|b|^2 + (|a|^2 - |b|^2)^2 = (|a|^2 + |b|^2)^2$. In the z chart it becomes

$$B([1 : z]) = \frac{(2\,\mathrm{Re}\, z,\ 2\,\mathrm{Im}\, z,\ 1 - |z|^2)}{1 + |z|^2}.$$

For a sphere point (X, Y, Z) away from the south pole, the inverse coordinate is $z = (X + iY)/(1 + Z)$. Away from the north pole it is $w = (X - iY)/(1 - Z)$. These formulas are inverse on their patches and agree by $w = 1/z$ on the overlap. Indeed, the displayed z formula for B is exactly the inverse south stereographic chart, with real coordinates $(\mathrm{Re}\, z, \mathrm{Im}\, z)$. On the w patch it is the inverse north chart with coordinates $(\mathrm{Re}\, w, -\mathrm{Im}\, w)$.

The already proved stereographic identities establish both inverse compositions. Hence B is a homeomorphism. In particular, $\mathbb{C}\mathrm{P}^1$ is Hausdorff and second-countable, inherited through this homeomorphism from the sphere. Its compatible charts therefore make it a smooth surface. The same formulas are smooth in both directions, so B is a diffeomorphism.

Take angles $0 \le \theta \le \pi$ and $0 \le \phi < 2\pi$. For the normalised representative $(a, b) = (\cos(\theta/2), e^{i\phi}\sin(\theta/2))$, the z coordinate, when $\theta < \pi$, is $z = \tan(\theta/2)e^{i\phi}$. At $\theta = \pi$ the state is $[0 : 1]$, with $w = 0$. The sphere image for the full angle range is $(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. The antipodal sphere point is represented by $(\sin(\theta/2), -e^{i\phi}\cos(\theta/2))$. Its inner product with the displayed representative is zero because

$$\cos(\theta/2)\sin(\theta/2) - \sin(\theta/2)\cos(\theta/2) = 0.$$

Thus antipodal Bloch vectors represent orthogonal pure states. The two-component state and the sphere point describe the same pure state, with no overall phase coordinate.

Self-check. Let $F\colon \mathbb{R}\mathrm{P}^1 \to \mathbb{R}\mathrm{P}^1$ be given by $F([X^0 : X^1]) = [X^0 + X^1 : X^0 - X^1]$. At $[2 : 3]$, compute its expression and derivative in the ratio coordinates $u = X^1/X^0$ on the source and $v = Y^1/Y^0$ on the target. State where this expression is valid and explain why F is smooth everywhere.

Answer. The matrix defining F has determinant -2, so a nonzero source never maps to zero. In the specified charts,

$$v = \frac{1-u}{1+u}, \qquad \frac{\mathrm{d}v}{\mathrm{d}u} = -\frac{2}{(1+u)^2}, \qquad u = \frac{3}{2}, \quad v = -\frac{1}{5}, \quad \left.\frac{\mathrm{d}v}{\mathrm{d}u}\right|_{u=3/2} = -\frac{8}{25}.$$

This expression is valid where $X^0 \neq 0$ and $X^0 + X^1 \neq 0$. In any source patch, choose a nonzero component of the target as denominator. The resulting coordinate expressions are ratios of linear functions with nonzero denominator on that neighbourhood, so they are smooth.

Self-check. State in one sentence what a smooth structure is and why it was introduced. Explain how the two structures on the line show that topology alone is insufficient.

Answer. A smooth structure is a maximal atlas of smoothly compatible charts, introduced to make smoothness of functions independent of coordinate descriptions. For $g(t) = t$, the expressions are s in the x chart and $\sqrt[3]{s}$ in the y chart. The second is not differentiable at zero, although both charts describe the same topological line.

Unit 7B Submanifolds and embeddings

7.5 Regular level sets and embeddings

We construct smooth manifolds as subsets of an ambient manifold. The local coordinates must also describe how the subset sits in that space.

Definition 7.16 (Embedded submanifold). A subset S of a smooth N-manifold M is an *embedded d-submanifold* if, with the subspace topology, it satisfies the following condition. At every $p \in S$ there is an ambient smooth chart (U, φ) containing p such that

$$\varphi(S \cap U) = \varphi(U) \cap (\mathbb{R}^d \times \{0\}).$$

The restricted coordinates give its smooth structure.

Definition 7.17 (Regular point)**.** For a smooth map $f\colon W \to \mathbb{R}^k$, where $W \subseteq \mathbb{R}^N$ is open, a point $p \in W$ is *regular* when Df_p is onto.

Definition 7.18 (Regular value)**.** For a smooth map $f\colon W \to \mathbb{R}^k$, a *regular value* is a point $c \in \mathbb{R}^k$ such that every point of $f^{-1}(c)$ is regular. A value outside the image is regular vacuously.

Surjectivity means that the constraint derivatives are independent. The hypothesis distinguishes two different failures. For $f(x,y) = xy$, the zero set is the union of the coordinate axes. Its derivative is the row (y,x), which vanishes at the crossing. Removing the crossing from a small connected neighbourhood leaves four components. An interval leaves two, and a punctured Euclidean ball of dimension at least two is connected. Thus the crossing has no Euclidean neighbourhood. For $f(x,y) = y^2$, the derivative is $(0, 2y)$. The value zero is again not regular, although its level set is the embedded x-axis. Regularity is sufficient, not necessary.

The graph $t = s^2$ becomes the slice $u = 0$ under $\Phi(s,t) = (s, t - s^2)$, as the drawing shows.

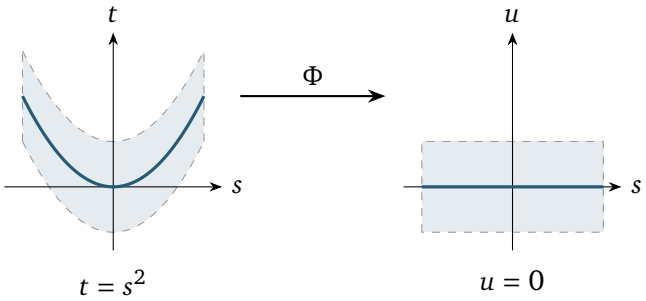


Figure 7.7: The open band $|s| < 1$, $|t - s^2| < 1/2$ maps to $|s| < 1$, $|u| < 1/2$. The graph becomes the horizontal slice.

Theorem 7.19 (Regular level sets · *proved given the implicit function theorem*)**.** *If c is a regular value of f, its nonempty level set $f^{-1}(c)$ is an embedded submanifold of W of dimension $N - k$.*

Proof. We use the implicit function theorem to write the level set as a graph, then subtract that graph in the constrained coordinates. Fix $p \in f^{-1}(c)$. Surjectivity gives a nonsingular $k \times k$ minor of Df_p. Permute the input coordinates and write them as $(s,t) \in \mathbb{R}^{N-k} \times \mathbb{R}^k$, with this minor in the t coordinates. The smooth implicit function theorem, cited in the form of Tu, Theorem B.2, gives open neighbourhoods in which

$$f(s,t) = c \quad \Longleftrightarrow \quad t = h(s)$$

for a smooth function h. In this neighbourhood the ambient coordinate change $\Phi(s,t) = (s, t - h(s))$ has smooth inverse $\Phi^{-1}(s,u) = (s, u + h(s))$. It sends the level set to $u = 0$. These are the required ambient charts. On their overlaps, the restricted transitions are smooth, because they are restrictions of smooth ambient transitions followed by projection onto the first $N - k$ components. The subspace is Hausdorff and second-countable, inherited from W. Thus these charts define the claimed embedded manifold. ∎

This is the Euclidean case of Tu, Theorem 9.9, pp. 105–106. For the sphere, use $f(x) = |x|^2$. At $p \in S^n$, its derivative is $Df_p(v) = 2p \cdot v$, which takes the value 1 on $v = p/2$. Hence 1 is

regular and the sphere has dimension $(n+1)-1=n$. The resulting structure agrees with the stereographic one, since the stereographic maps and their inverses are smooth ambient formulas on their respective domains.

For the orthogonal group, the constraint takes values in the vector space Sym_n of real symmetric matrices:

$$f\colon \mathbb{R}^{n\times n} \to \mathrm{Sym}_n, \qquad f(A) = A^T A.$$

The target has dimension $n(n+1)/2$, since its diagonal and entries above the diagonal determine it. Expanding $f(A+tH)$ to first order in the real parameter t gives

$$Df_A(H) = A^T H + H^T A.$$

At $A \in \mathrm{O}(n)$, any symmetric matrix S is the image of $H = AS/2$, since $A^T A = I$. Thus I is a regular value and $\dim \mathrm{O}(n) = n^2 - n(n+1)/2 = n(n-1)/2$. Taking all n^2 matrix entries as independent target constraints would incorrectly count the repeated off-diagonal equations.

Definition 7.20 (Immersion)**.** A smooth map is an *immersion* if its coordinate Jacobian is injective at every point.

Definition 7.21 (Embedding)**.** An immersion is an *embedding* if it is also a homeomorphism onto its image with the subspace topology.

Chart changes multiply the Jacobian by invertible matrices, so the immersion condition is independent of charts.

The inclusion of an embedded submanifold is an embedding. In slice charts its Jacobian is the inclusion of $\mathbb{R}^d$ into $\mathbb{R}^N$. Its topology is the subspace topology by definition.

Conversely, the image of an embedding is an embedded submanifold. Indeed, a rank-d minor makes d target coordinates local coordinates on the source, by the inverse function theorem. In these coordinates the remaining components form a smooth graph. Subtracting that graph gives a slice chart. The homeomorphism onto the image lets us restrict the ambient neighbourhood so that no other part of the image enters this local graph.

For comparison, the cusp parametrisation $t \mapsto (t^3, t^2)$ has zero velocity at $t=0$, so it is not an immersion. The curve $t \mapsto (\sin t, \sin 2t)$, with t taken modulo 2π, is an immersion: if $\cos t = 0$, then $\cos 2t = -1$. It is not injective, since $t=0$ and $t=\pi$ both map to $(0,0)$. The circle inclusion is an embedding. The drawing compares these maps with the irrational curve considered next.

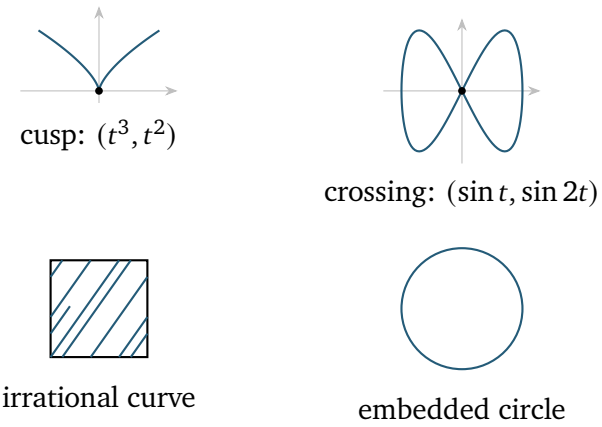


Figure 7.8: The cusp fails the derivative test, and the crossing curve fails injectivity. The irrational curve is injective but fails the subspace-topology test proved below.

An injective immersion need not be an embedding. Fix an irrational number $\alpha > 0$ and consider

$$\gamma : \mathbb{R} \longrightarrow T^2, \qquad \gamma(t) = (e^{it}, e^{i\alpha t}).$$

Equal images require a difference $2\pi m$ with $m \in \mathbb{Z}$ and $\alpha m \in \mathbb{Z}$. Irrationality forces $m = 0$, proving injectivity. The first circle component has nonzero derivative, proving immersion.

We test continuity of the inverse at $\gamma(0)$ by finding large parameters whose images approach $\gamma(0)$. At times $t = 2\pi q$ with a positive integer q, the first component is exactly 1. The second is $e^{2\pi i \alpha q}$, so we seek αq close to an integer. The square model uses coordinates $t/(2\pi)$ and $\alpha t/(2\pi)$ modulo integers. It shows a finite part of these repeated returns.

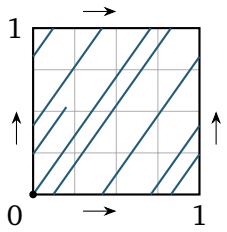


Figure 7.9: A finite segment of the line of slope $\sqrt{2}$ in the square quotient $\mathbb{R}^2/\mathbb{Z}^2$. Each exit continues at the identified opposite edge.

We produce unbounded positive integers q_j whose distances from αq_j to the integers tend to zero. Divide $[0, 1)$ into N equal half-open intervals. Place the $N + 1$ fractional parts of $0, \alpha, \ldots, N\alpha$ into them. Two lie in the same interval. Subtracting their indices gives $1 \le q \le N$ with distance from αq to an integer less than $1/N$. If these choices of q stayed bounded as N increased, one positive integer q would recur for arbitrarily large N. Its fixed distance to the integers would then be zero. This would give $\alpha q \in \mathbb{Z}$, contradicting irrationality. Hence we can extract an unbounded subsequence of choices with $N \to \infty$. Consequently $\gamma(2\pi q_j) \to \gamma(0)$, although $2\pi q_j$ does not tend to zero. The inverse from the image is not continuous. Thus γ is not an embedding. Lecture 9 describes motion along this curve by prescribing its velocity at every point.

For $\alpha = \sqrt{2}$, the returns at $t = 10\pi$ and $t = 58\pi$ have local coordinates $(0, 5\sqrt{2} - 7)$ and $(0, 29\sqrt{2} - 41)$. The drawing compares nearby image points with their separated parameters.

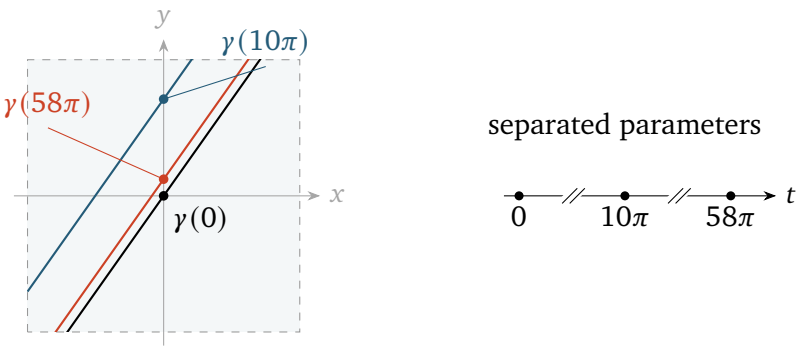


Figure 7.10: In local torus coordinates, return strands have equation $y = \sqrt{2}x + q\sqrt{2} - p$ for $(q, p) = (0, 0), (5, 7), (29, 41)$. The parameter axis uses breaks between the labelled values.

Self-check. Use the regular level set theorem to show that $\mathrm{U}(n)$ is a real manifold of dimension n^2. Specify the real target of the constraint. For the following unitary matrix A and Hermitian matrix S, exhibit H with $Df_A(H) = S$:

$$A = \begin{pmatrix} 3/5 & 4\mathrm{i}/5 \\ 4\mathrm{i}/5 & 3/5 \end{pmatrix}, \qquad S = \begin{pmatrix} 2 & 1+\mathrm{i} \\ 1-\mathrm{i} & -1 \end{pmatrix}.$$

Answer. Use $f(A) = A^\dagger A$ on $\mathbb{C}^{n\times n} \cong \mathbb{R}^{2n^2}$ with target the real vector space of Hermitian matrices. That target has n real diagonal entries and one complex entry for each pair above the diagonal, so its real dimension is $n + 2\binom{n}{2} = n^2$. Differentiating with respect to a real parameter gives $Df_A(H) = A^\dagger H + H^\dagger A$. At a unitary A, every Hermitian S is reached by $H = AS/2$. Thus I is a regular value and its level set has dimension $2n^2 - n^2 = n^2$. Treating this as a complex-linear constraint would be incorrect: conjugation is part of the formula. For the specified data, the same right inverse gives

$$H = \frac{AS}{2} = \begin{pmatrix} 1+2\mathrm{i}/5 & 3/10-\mathrm{i}/10 \\ 3/10+\mathrm{i}/2 & -7/10+2\mathrm{i}/5 \end{pmatrix}.$$

Substitution into $A^\dagger H + H^\dagger A$ returns S.

Summary. A smooth structure is specified by charts with smooth transitions. Smooth maps are tested in these charts, and the Euclidean chain rule makes the test independent of coordinates. The implicit function theorem turns independent smooth constraints into embedded submanifolds. The qubit state space has the two projective charts z and $w = 1/z$, and is diffeomorphic to the Bloch sphere. Lecture 8 uses coordinate changes to define tangent vectors and differentials intrinsically.

Reading. Tu, *An Introduction to Manifolds*, §§5–7, Theorem 9.9 and §11; the irrational-line example is Example 15.9. Nakahara, *Geometry, Topology and Physics*, §5.1.